Zomato Dataset Analysis Report

Alfido Tech Internship – Task 1

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1. Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Zomato dataset to identify trends, customer preferences, restaurant ratings, pricing patterns, and online delivery services.

2. Dataset Description

The dataset contains information related to restaurants listed on Zomato. It includes restaurant names, ratings, votes, location details, online delivery availability, table booking options, and approximate cost for two people

3. Data Cleaning and Preprocessing

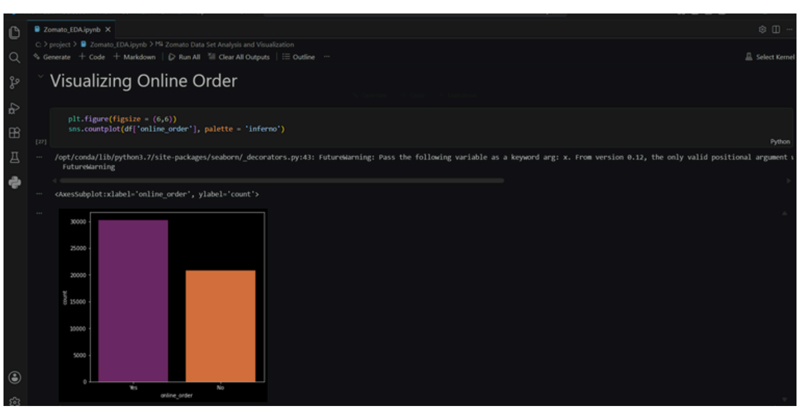
Several preprocessing steps were performed before analysis:

- Missing values were identified and handled.
- Duplicate records were checked and removed if necessary.
- Data types were verified and corrected.
- Irrelevant columns were removed for better analysis.
- Column names were standardized for readability.

4. Exploratory Data Analysis

Different visualization techniques were used to understand the dataset:

- Rating distribution analysis
- Online delivery analysis
- Cost distribution analysis
- Restaurant type analysis
- Votes and customer engagement trends



No. of Votes, Location Wise

```
df4 = df[['location', 'votes']]
df4.drop_duplicates()
df5 = df4.groupby(['location'])['votes'].sum()
df5 = df5.to_frame()
df5 = df5.sort_values('votes', ascending=False)
df5.head()
```

[38]

location	votes
Koramangala 5th Block	2214083
Indiranagar	1165909
Koramangala 4th Block	685156
Church Street	590306
JP Nagar	586522

```
plt.figure(figsize = (15,8))
sns.barplot(df5.index , df5['votes'])
plt.xticks(rotation = 90)
```

[39]

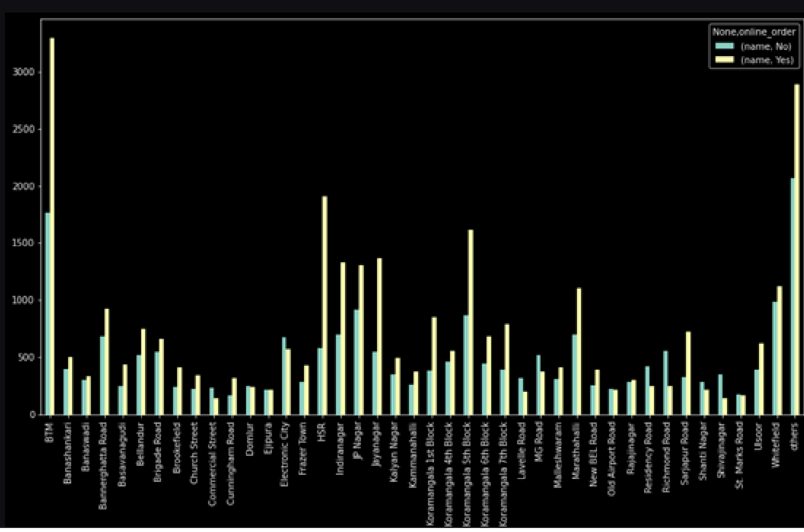
```
/opt/conda/lib/python3.7/site-packages/seaborn/_decorators.py:43: FutureWarning: Pass the following variables as keyword args: x, y. From version 0.12, the only valid positional ar
  FutureWarning

(array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
        17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
        34, 35, 36, 37, 38, 39, 40, 41]),
```

```
df1.plot(kind = 'bar', figsize = (15,8))
```

[32]

```
<AxesSubplot:xlabel='location'>
```



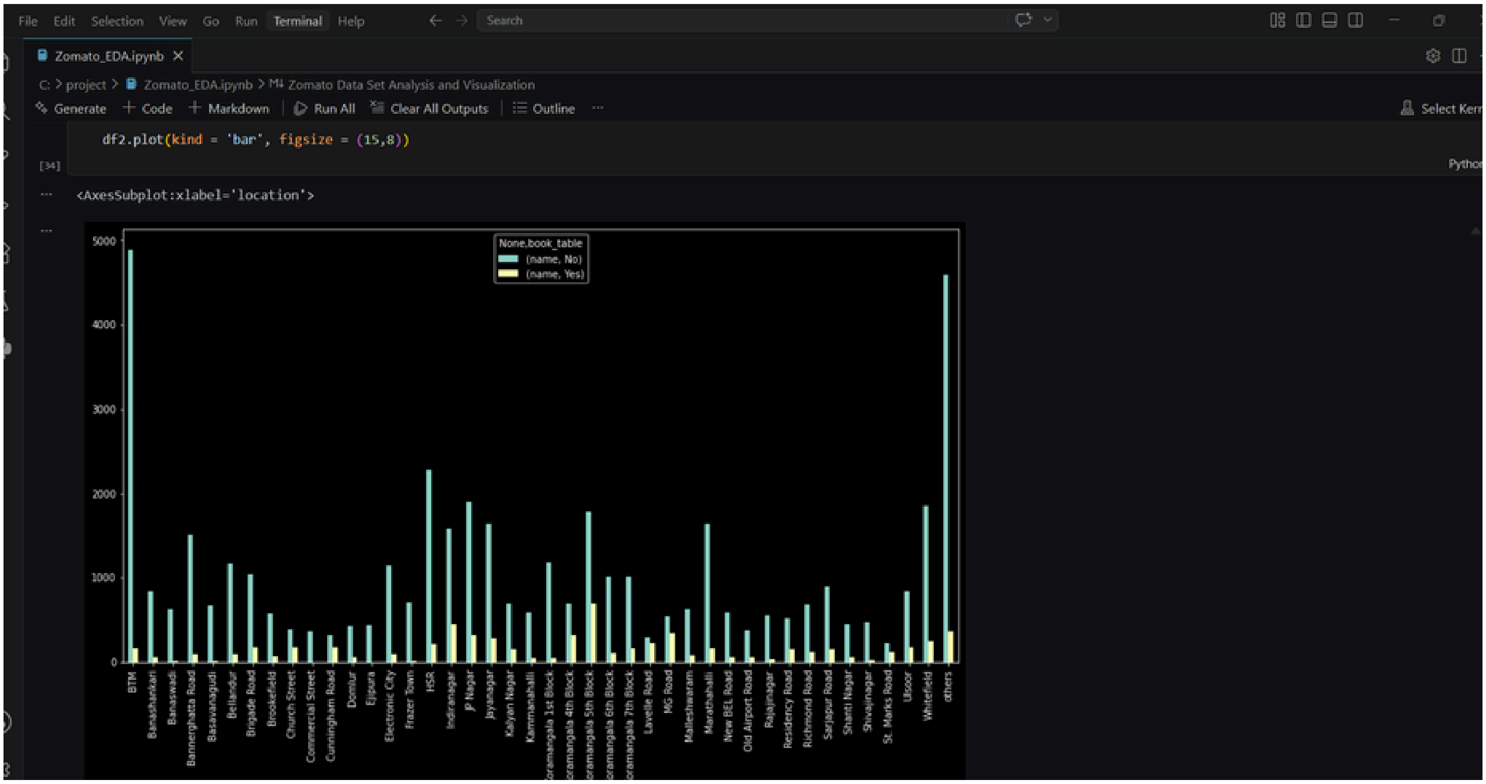

```
df.head()
```

[40]

	name	online_order	book_table	rate	votes	location	rest_type	cuisines	Cost2plates	Type
0	Jalsa	Yes	Yes	4.1	775	Banashankari	Casual Dining	North Indian, Mughlai, Chinese	800.0	Buffet
1	Spice Elephant	Yes	No	4.1	787	Banashankari	Casual Dining	others	800.0	Buffet
2	San Churro Cafe	Yes	No	3.8	918	Banashankari	others	others	800.0	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	Banashankari	Quick Bites	South Indian, North Indian	300.0	Buffet
4	Grand Village	No	No	3.8	166	Basavanagudi	Casual Dining	others	600.0	Buffet

Visualizing Top Cuisines

```
df6 = df[['cuisines', 'votes']]
df6.drop_duplicates()
df7 = df6.groupby(['cuisines'])['votes'].sum()
df7 = df7.to_frame()
df7 = df7.sort_values('votes', ascending=False)
```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Zomato_EDA.ipynb
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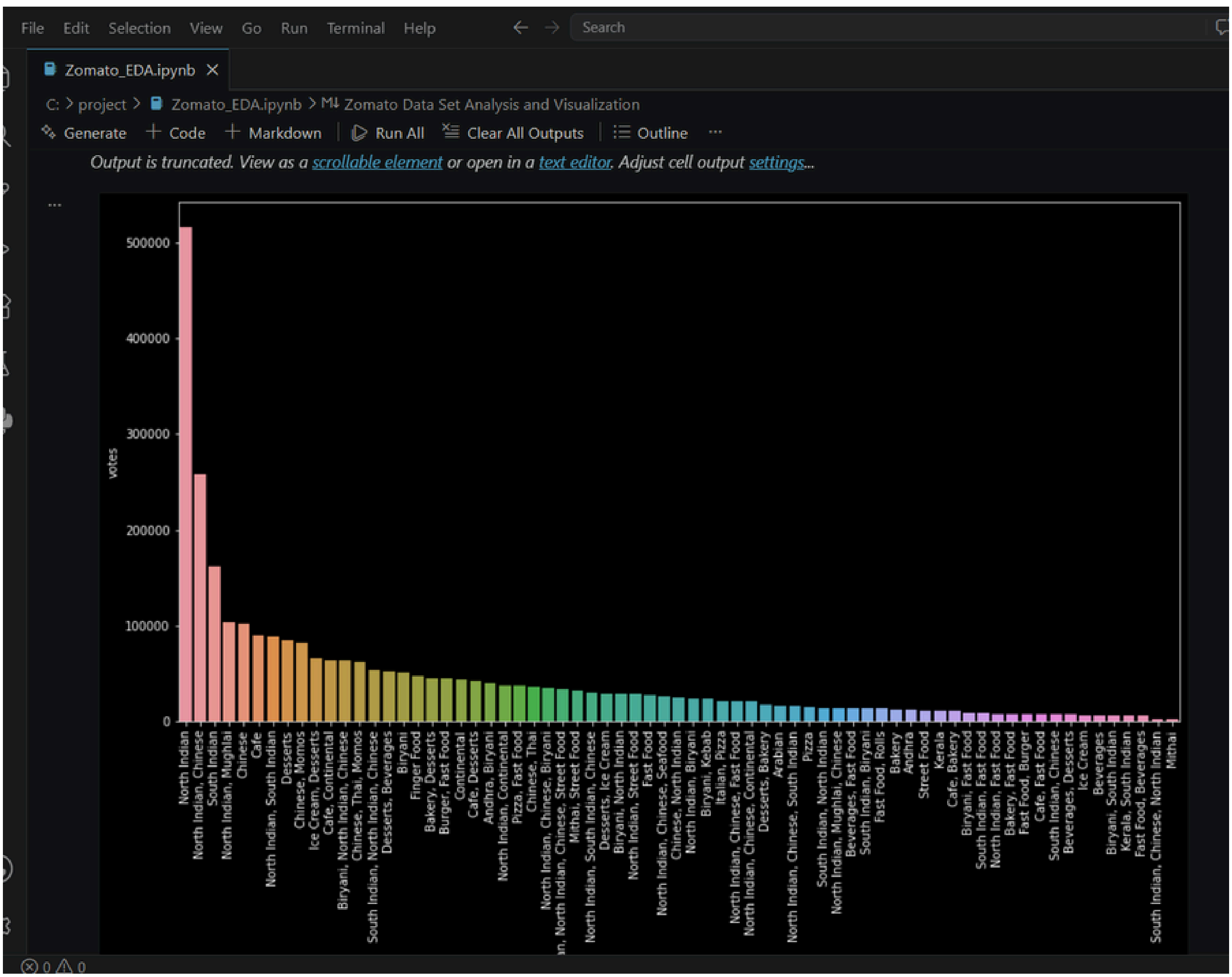
Generate + Code + Markdown Run All Clear All Outputs Outline
df2.plot(kind = 'bar', figsize = (15,8))
[34]
<AxesSubplot:xlabel='location'>
None,book_table
(name, No)
(name, Yes)
5000
4000
3000
2000
1000
0
BTM
Banashankari
Banaswadi
Bannerghatta Road
Basavanagudi
Bellandur
Brigade Road
Brookefield
Church Street
Commercial Street
Cunningham Road
Domlur
Ejipura
Electronic City
Frazer Town
HSR
Indiranagar
JP Nagar
Jayanagar
Kalyan Nagar
Kammanahalli
Lavelle Road
MG Road
Malleshwaram
Marathahalli
New BEL Road
Old Airport Road
Rajajinagar
Residency Road
Richmond Road
Sarjapur Road
Shanti Nagar
Shivajinagar
St. Marks Road
Ulsoor
Whitefield
others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Search
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Generate + Code + Markdown Run All Clear All Outputs Outline
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
votes
500000
400000
300000
200000
100000
0
North Indian
North Indian, Chinese
South Indian
North Indian, Mughlai
Chinese
Cafe
North Indian, South Indian
Desserts
Chinese, Momos
Ice Cream, Desserts
Cafe, Continental
Biryani, North Indian, Chinese
Chinese, Thai, Momos
South Indian, North Indian, Chinese
Desserts, Beverages
Biryani
Finger Food
Bakery, Desserts
Burger, Fast Food
Continental
Cafe, Desserts
Andhra, Biryani
North Indian, Continental
Pizza, Fast Food
Chinese, Thai
North Indian, Chinese, Biryani
North Indian, Chinese, Street Food
Mithai, Street Food
North Indian, South Indian, Chinese
Desserts, Ice Cream
Biryani, North Indian
North Indian, Street Food
Fast Food
North Indian, Chinese, Seafood
Chinese, North Indian
North Indian, Biryani
Biryani, Kebab
Italian, Pizza
North Indian, Chinese, Fast Food
North Indian, Chinese, Continental
Desserts, Bakery
Arabian
North Indian, Chinese, South Indian
Pizza
South Indian, North Indian, Chinese
North Indian, Mughlai, Chinese
Beverages, Fast Food
South Indian, Biryani
Fast Food, Rolls
Bakery
Andhra
Street Food
Kerala
Cafe, Bakery
Biryani, Fast Food
South Indian, Fast Food
North Indian, Fast Food
Bakery, Fast Food
Fast Food, Burger
Cafe, Fast Food
South Indian, Chinese
Beverages, Desserts
Ice Cream
Beverages
Biryani, South Indian
Kerala, South Indian
Fast Food, Beverages
South Indian, Chinese, North Indian
Mithai
⊗ 0 ⚠ 0

5. Key Insights

• Most restaurants have ratings between 3.5 and 4.0.
• Online delivery services are widely used among restaurants.

• Mid-range restaurants are more common compared to premium restaurants.
• Restaurants with higher ratings generally receive more customer votes.
• Casual dining and cafe categories dominate the dataset.

6. Conclusion

The analysis provided meaningful insights into restaurant trends and customer preferences using the Zomato dataset. EDA techniques and visualizations helped identify important business patterns related to ratings, pricing, and online delivery.